

National Park and Wildlife Sanctuaries

- There are 103 National Parks and 528 Wildlife Sanctuaries covering an area of more than 15.67 million ha in the country.
- Madhya Pradesh and Andaman and Nicobar Islands have the maximum number of National Parks (9 each).
- Andaman and Nicobar Islands has 96 (maximum in India) and Maharashtra has 36 wildlife sanctuaries.

Difference between National Park, Sanctuary and Biosphere Reserve

National Park	Sanctuary	Biosphere Reserve
A reserved area for preservation of its natural vegetation, wildlife and natural beauty.	A reserved area for preservation of endangered species.	Multipurpose protected area to preserve genetic diversity in representative ecosystem.
Boundaries are fixed by legislation.	Boundaries are not sacrosanct.	Boundaries are fixed by legislation.

Important Sanctuaries and National Parks

Name	Location	Reserve For
Achanakmar Sanctuary	Chhattisgarh	Tiger, boar, cheetal, sambhar and bison
Bandhavgarh National Park	Madhya Pradesh	Tiger, panther, cheetal, nilgai and wild boar
Bandipur Sanctuary	Karnataka and Tamil Nadu	Elephant, tiger, panther, sambhar, deer and birds
Banerghatta National Park	Karnataka	Elephant, cheetal, deer and grey partridge and green pigeon
Bhadra Sanctuary	Karnataka	Elephant, cheetal, panther, sambhar and wild boar
Chandraprabha Sanctuary	Uttar Pradesh	Gir lions, cheetal and sambhar
Corbett National Park	Uttarakhand	Tiger, leopard, elephant and sambhar (named in memory of Jim Corbett)
Dachigam Sanctuary	Jammu and Kashmir	Kashmiri stag
Dandeli Sanctuary	Karnataka	Tiger, panther, elephant, cheetal, sambhar and wild boar
Dudhwa National Park	Uttar Pradesh	Tiger, panther, sambhar, cheetal, nilgai and barking deer
Gandhi Sagar Sanctuary	Madhya Pradesh	Cheetal, sambhar, chinkara and wild birds
Ghana Bird Sanctuary	Rajasthan	Water birds, black-buck, cheetal and sambhar
Gir Forest	Gujarat	India's biggest wildlife sanctuary famous for gir lions
Gautam Buddha Sanctuary	Bihar	Tiger, leopard, sambhar, cheetal and barking deer
Jaldapara Sanctuary	West Bengal	Rhinoceros

Kaziranga National Park	Assam	One horned rhinoceros, gaur, elephant, leopard and wild buffalo
Kanchenjunga National Park	Sikkim	Snow leopard, musk deer and Himalayan boar
Nagarhole National Park	Karnataka	
Namdapha Sanctuary	Arunachal Pradesh	Elephant, panther, sambhar, tiger, cheetal and king cobra
Pachmarhi Sanctuary	Madhya Pradesh	Tiger, panther, boar, sambhar, nilgai and barking deer
Simlipal Sanctuary	Odisha	Elephant, tiger, leopard, gaur and cheetal
Sunderban Tiger Reserve	West Bengal	Tiger, deer, wild boar, crocodile and Gangetic dolphin
Sonai Rupa Sanctuary	Assam	Elephant, sambhar, wild boar and one-horned rhinoceros
Tungabhadra Sanctuary	Karnataka	Panther, cheetal, sloth bear and four-horned antelope
Valadore National Park	Gujarat	Wolf and black buck
Vedanthangal Bird Sanctuary	Tamil Nadu	Important bird sanctuary
Wild Ass Sanctuary	Gujarat	Wild ass, wolf, nilgai and chinkara

Biosphere Reserves in India

- The biosphere reserve programme was launched by the UNESCO in 1971, under the aegis of its Man and Biosphere (M&B) programme, to provide a global network of protected areas for conserving natural communities.
- There are 18 biosphere reserves in India of which of 10 recognised by UNESCO.

Biosphere Reserves of India

Name	States	Type	Area (km²)
Great Rann of Kutch	Gujarat	Desert	12454
Gulf of Mannar (UNESCO)	Tamil Nadu	Coasts	10500
Sunderbans (UNESCO)	West Bengal	Gangetic Delta	9630
Cold Desert	Himachal Pradesh	Western Himalayas	7770
Nanda Devi (UNESCO)	Uttarakhand	West Himalayas	5860
Nilgiri (UNESCO)	Tamil Nadu, Kerala and Karnataka	Western Ghats	5520
Dihang-Dibang	Arunachal Pradesh	East Himalayas	5112
Pachmarhi (UNESCO)	Madhya Pradesh	Semi-Arid	4926
Seshachalam Hills	Andhra Pradesh	Eastern Ghats	4755.997
Simlipal (UNESCO)	Odisha	Deccan Peninsula	4374
Achanakmar-Amarkantak (UNESCO)	Madhya Pradesh, Chhattisgarh	Maikala Range	3835

Name	States	Type	Area (km ²)
Manas	Assam	East Himalayas	2837
Khanchenjunga	Sikkim	East Himalayas	2620
Agasthyamalai (UNESCO)	Kerala, Tamil Nadu	Western Ghats	1828
Great Nicobar (UNESCO)	Andaman and Nicobar Islands	Islands	885
Dibru-Saikhowa	Assam	East Himalayas	765
Panna	Madhya Pradesh	Catchment area of Ken river	2998
Nokrek (UNESCO)	Meghalaya	East Himalayas	48

CONTEMPORARY ENVIRONMENTAL ISSUES

Climate Change

- Changes in the average weather for a particular location leads to climate change. It result from both natural processes such as the change in the Sun's strength and also from human activities, through the build-up of greenhouse gases.
- The atmospheric concentrations of these gases have increased significantly, since, pre-industrial times largely because of fossil fuel usage, decrease in forest cover etc resulting in climate change.
- It is now a global concern that the climatic changes occurring today have been speeded up because of man's activities.

Causes of Climate Change

The causes of climate change can be divided into two categories, which are as follow :

- Natural Causes** The Earth's climate is influenced and changed through natural causes like volcanic eruptions, ocean current, the Earth's orbital changes and solar variations.
- Man-Made Causes** Burning of fossil fuels, deforestation, mining, industrialisation etc are man-made causes of climate change.

Effects of Climate Change

- Climate changes can severely affect human societies, agriculture and the natural ecosystem, terrestrial and aquatic ecosystem, which provide many goods and services on which we rely, would be severely affected.
- Combined with oxidation of high latitude peat lands, release of the carbon stores would add greatly to the CO₂ content in the atmosphere. Thus, the effects of warming themselves would cause more warming.
- Many wild plants and animal species found today can be forced out of their present area of growth/habitats as climate warms.

- Climate change would impact on agricultural and forestry management.
- Rising sea levels and heavy storm damage would severely affect coastlines. Sea levels are increasing at about 2 mm per year. A 1m rise in sea level over the 21st century would mean sub-mergence of some low-lying island nations and displacement of a large number of people.
- Infectious disease would become more common as global temperatures rise.

Global Warming

- This refers to an increase in average global temperature. Natural events and human activities are believed to be contributing to an increase in average global temperature.
- This is caused primarily by increases in 'greenhouse' gases such as carbon dioxide (CO₂), methane nitrous oxide etc.

Greenhouse Effect

- The term 'greenhouse' is used in conjunction with the phenomenon known as the Greenhouse Effect.
- Six main greenhouse gases are Carbon Dioxide (CO₂), Methane (CH₄) (which is 20 times as potent a greenhouse gas as carbon dioxide) and Nitrous Oxide (N₂O), plus three fluorinated industrial gases: Hydrofluorocarbons (HFCs), Perfluorocarbons (PFCs) and Sulphur Hexafluoride (SF₆).

Impact of Global Warming

Extreme Weather Patterns Most scientists believe that the warming of the climate will lead to more extreme weather patterns. Such as, more hurricanes and drought, longer spells of dry heat or intense rain (depending on where you are in the world).

Rising Sea Levels Water expands when heated and sea levels are expected to rise due to climate change. Rising sea levels will also result as the polar caps begin to melt. Rising sea levels is already affecting many small Islands.

Ocean Acidification Oceans are able to absorb some of the excess CO₂ released by human activity. This has helped to keep the planet cooler. But, due to it pH level of water become low, which affects marine life especially to coral and animals having calcium shell.

Global Dimming Inter-Governmental Panel on Climate Change (IPCC). This is the leading international body for the assessment of climate change. It was established by the United Nations Environment Programme (UNEP) and the World Meteorological Organisation (WMO) in 1988 to provide the world with a clear scientific view on the current state of knowledge in climate change and its potential environmental and socio-economic impacts.

United Nations Framework Convention on Climate Change (UNFCCC)

- The United Nations Framework Convention on Climate Change (UNFCCC), signed by over 150 countries at the Rio Earth Summit in 1992. *The main purposes of this protocol was to*
 - provide mandatory targets on greenhouse gas emissions for the world's leading economies, all of whom accepted it at the time.
 - provide flexibility in how countries meet their targets.
 - further recognise that commitments under the protocol would vary from country to country.

Kyoto Protocol

- The Kyoto Protocol, 1997 to the UNFCCC is an international treaty that sets binding obligations on industrialised countries to reduce emissions of greenhouse gases.
- The protocol recognises that developed countries are principally responsible for the current high levels of greenhouse gas emissions in the atmosphere, as a result of more than 150 years of industrial activity and places a heavier burden on developed nations under the principle of common, but differentiated responsibilities.

Paris Agreement

- It is dealing with Green House Gas (GHG) emissions mitigation, adaptation and finance from year 2020.
- This agreement has been signed by 191 countries and 61 countries have even ratified it. This agreement will come into force when 55 countries that emits 55% GHG ratify it.

Hole in Ozone Layer

- The ozone layer lies in the stratosphere, in the upper level of our atmosphere. The ozone in it is spread very sparsely.
- Stratospheric ozone filters out most of the Sun's potentially harmful shortwave Ultraviolet (UV) radiation.
- This ozone has become depleted, due to the release of such ozone-depleting substances such as Chlorofluorocarbons (CFCs). When stratospheric ozone is depleted, more UV rays reach the Earth.
- Exposure to higher amounts of UV radiation could have serious impacts on human beings, animals and plants.

Depletion of the Stratospheric Ozone Layer (Ozone Depletion)

- In 1985, a group of scientists made an unsettling discovery, a marked decrease in stratospheric ozone over the South pole, in the Antarctic. The depletion

appeared during the Southern hemisphere's spring (October and November) and then filled in. Soon after the Antarctic hole was found, Canadian scientists discovered that the ozone layer above the Arctic is also thinning significantly.

- The highest latitudes; the North and South poles experience the greatest amount of ozone loss, during their spring. Ozone depletion is most pronounced in the Antarctic. But ozone depletion, to a lesser degree, now occurs in the mid latitudes. e.g. the amount of stratospheric ozone over the Northern hemisphere has been dropping by 4% per decade.

Impacts of Ozone Layer Depletion

- Stratospheric ozone filters out most of the Sun's potentially harmful shortwave Ultraviolet (UV) radiation. If this ozone becomes depleted, then more UV rays will reach the Earth. Exposure to higher amounts of UV radiation could have serious impacts on human beings, animals and plants, such as the following
 - **Impact on Human Life** More skin cancers, sunburns and premature aging of the skin. UV radiation can damage several parts of the eye, including the lens, cornea, retina and conjunctiva. Cataracts (a clouding of the lens) are the major cause of blindness in the world.
 - **Agriculture, Forestry and Natural Ecosystems** Several of the world's major crop species are particularly vulnerable to increased UV, resulting in reduced growth, photosynthesis and flowering.
 - These species include wheat, rice, barley, oats, corn, soyabeans, peas, tomatoes, cucumbers, cauliflower, broccoli and carrots.
 - **Damage to Marine Life** In particular, plankton (tiny organisms in the surface layer of oceans) are threatened by increased UV radiation. Decreases in plankton could disrupt the fresh and saltwater food chains and lead to a species shift in canadian waters. Loss of biodiversity in our oceans, rivers and lakes could reduce fish yields for commercial and sport-fisheries.
 - **Impacts on Animals** In domestic animals, UV over exposure may cause eye and skin cancers. Species of marine animals in their development stage (e.g. young fish, shrimp larvae and crab larvae) have been threatened in recent years by the increased UV radiation under the Antarctic ozone hole.
 - **Impacts on Materials** Wood, plastic, rubber, fabrics and many construction materials are degraded by UV radiation. The economic impact of replacing and/or protecting materials could be significant.

The Main Ozone Depleting Substances (ODSs)

The main ozone of depleting substances are as follow :

- **Chlorofluorocarbons (CFCs)** The most widely used ODS, accounting for over 80% of total stratospheric ozone depletion. Used as coolants in refrigerators, freezers and air conditioners in buildings and cars manufactured before 1995.

Found in industrial solvents, dry cleaning agents and hospital sterilants. Also used in foam products such as soft-foam padding (e.g. cushions and mattresses) and rigid foam (e.g. home insulation).

- **Halons** Used in some fire extinguishers, in cases where materials and equipment would be destroyed by water or other fire extinguisher chemicals.
- **Methyl Chloroform** Used mainly in industry for vapour decreasing, some aerosols, cold cleaning, adhesives and chemical processing.
- **Carbon Tetrachloride** Used in solvents and some fire extinguishers.
- **Hydrochlorofluoro Carbons (HCFCs)** HCFCs have become major, transitional substitutes for CFCs. They are much less harmful to stratospheric ozone than CFCs are, but HCFCs, still cause some ozone destruction and are potent greenhouse gases.

Montreal Protocol

- Ozone depletion has generated worldwide concern leading to the adoption of montreal protocol which was signed in 1987. This protocol bans the production of CFCs, halons and other ozone depleting chemicals such as carbon tetrachloride and trichloroethane.
- Since 1987, more than 150 countries have signed an international agreement on the montreal protocol, which called for a phased reduction in the release of CFCs, such that the yearly amount added to the atmosphere in 1999, would be half that of 1986. Modifications of that treaty called for a complete ban on CFCs, which began in January, 1996.

Environment Related Important International Agreements/Conference

UN Conference on the Human Environment	Stockholm (1972)
Convention on Migratory Species	Bonn (1979)
Convention for the Protection of the Ozone Layer	Vienna (1985)
Protocol on Substances that Deplete the Ozone Layer	Montreal (1987)
Convention on the Transboundary Movement of Hazardous Wastes	Basel (1989)
Earth Summit (<i>UN Conference on Environment and Development</i>)	Rio-de-Janeiro (1992)
Convention on Prior Informed Consent	Rotterdam (1998)
UN Conference on Sustainable Development	Rio-de-Janeiro (2012)
Nagoya Protocol on Genetic Resources	Nagoya (2010)
Convention on Biological Diversity (CBD-CoP-11)	Hyderabad (2012)
UN Climate Change Conference (CoP-20)	Lima (2014)
Paris Climate Conference (CoP-21)	Paris (2015)

National Action Plan on Climate Change (NAPCC)

National Action Plan on Climate Change (NAPCC) is a comprehensive action plan which outlines measures on climate change related adaptation and mitigation while simultaneously advancing development. The eight missions

from the core of the plan, representing multi-pronged, long termed and integrated strategies for achieving goals in the context of climate change. *The eight missions are as follow :*

- National Solar Mission** The NAPCC aims to promote the development and use of solar energy for power generation and other uses, with the ultimate objective of making solar competitive with fossil-based energy options.
- National Mission for Enhanced Energy Efficiency** The NAPCC recommends mandating specific energy consumption decreases in large energy-consuming industries, with a system for companies to trade energy-saving certificates, financing for public-private partnerships to reduce energy consumption.
- National Mission on Sustainable Habitat** The NAPCC also aims at promoting energy efficiency as a core component of urban planning by extending the existing Energy Conservation Building Code, strengthening the enforcement of automotive fuel economy standards, and using pricing measures to encourage the purchase of efficient vehicles and incentives for the use of public transportation.
- National Water Mission** The NAPCC sets a goal of a 20% improvement in water use efficiency through pricing and other measures to deal with water scarcity as a result of climate change.
- National Mission for Sustaining the Himalayan Ecosystem** This particular mission sets the goal to prevent melting of the Himalayan glaciers and to protect biodiversity in the Himalayan region.
- Green India Mission** The NAPCC also aims at afforestation of 6 million hectares of degraded forest lands and expanding forest cover from 23 to 33% of India's territory.
- National Mission for Sustainable Agriculture** The NAPCC aims to support climate adaptation in agriculture through the development of climate-resilient crops, expansion of weather insurance mechanisms and agricultural practices.
- National Mission on Strategic Knowledge for Climate Change** To gain a better understanding of climate science, impacts, and challenges, the plan envisions a new Climate Science Research Fund, improved climate modeling, and increased international collaboration.

Sustainable Development

- Sustainable development is the development that meets the need of the present generation without compromising the ability of future generations to meet their needs.
- It is an organising principle for human life on a finite planet. It put forward or desirable future state for human societies, in which living conditions and resource use meet human needs without undermining the sustainability of natural systems and environment.

> PRACTICE EXERCISE

1. Match the following

List I (Green House Gases)	List II (Major Sources)
A. Carbon dioxide	1. Enteric fermentation in cattle
B. Methane	2. Burning of fossil fuels
C. Chlorofluoro carbons	3. Mercury compounds in waste water
D. Nitrous oxide	4. Nylon production
	5. Air-conditioners and refrigeration units

Codes

A B C D	A B C D
(a) 2 1 5 4	(b) 3 2 4 1
(c) 3 4 2 1	(d) 1 2 3 5

2. What is the primary reason for targeting 'biodiversity hotspots' for conservation?

- To protect all areas of threatened species would not allow for new species to develop
- Because the hotspots are Earth-biologically richest and most endangered terrestrial eco-regions
- They are areas where people do not live and conservation would therefore not be effecting the economic development of the area
- They are the only areas where species are seriously threatened in the world

3. The United Nations Framework Conventions on Climate Change (UNFCCC) is an international treaty drawn at

- United Nations Conference on the Human Environment, Stockholm, 1972
- UN Conference on Environment and Development, Rio de Janeiro, 1992
- World Summit on Sustainable Development, Johannesburg, 2002
- UN Climate Change Conference Copenhagen, 2009

4. The Chinkara (*Gazella Bennettii*) or Indian Gazelle is found in

- Rajasthan, North-Western and central parts of India
- North-East
- Northernmost Himalayas
- South India

5. Which one of the following pairs is not correctly matched?

National Park	State
(a) Kanha National Park	: Madhya Pradesh
(b) Sultanpur National Park	: Haryana
(c) Ranthambhore National Park	: Gujarat
(d) Bandipur National Park	: Karnataka

6. In the context of ecosystem productivity, marine upwelling zones are important as they increases the marine productivity by bringing the

1. decomposer micro-organisms to the surface.
2. nutrient to the surface.
3. bottom dwelling organisms to the surface.

Which of the statement(s) given above is/are correct?

- 1 and 2
- Only 2
- 2 and 3
- Only 3

7. Which of the following air (s) of projects are related with environmental problems?

1. River Valley Project : Water body induced
2. Mining Project : Groundwater and surface water pollution
3. Thermal power project : Degradation of land

Which of the pair(s) given above is/are correctly matched?

- Only 1
- 2 and 3
- 1 and 2
- All of these

8. Which of the following is not a major positive feedback mechanism in which the activity of humans to increase global climate temperature leads to an even further increases?

- Tropical deforestation causes warming and drying so that remaining forests begin to decline
- Global warming causes snow to melt in polar regions and therefore increases global albedo
- Global warming causes increased rainfall, plant growth and photosynthesis
- Global warming causes increased CO₂ release from biomass decomposition

9. Proper disposal of hazardous toxic waste is essential as exposure to it can causes serious problems to the health. Which is potentially the safest means of disposing of the most toxic wastes: organic solvents, chlorinated hydrocarbons and oil-base compounds (including PCBs are dioxins)?

- Municipal incineration
- Industrial high temperature incineration
- Landfills
- None of the above

10. Consider the following statement(s)

1. The boundaries of a National Park are defined by legislation.
2. A biosphere reserve is declared to conserve a few specific species of flora and fauna.
3. In a wildlife sanctuary, limited biotic interference is permitted.

Which of the statement(s) given above is/are correct?

- Only 1
- 2 and 3
- 1 and 3
- All of these

11. Which among the following is the most abundant Green House Gas (GHG) in the Earth's atmosphere?

- Carbon dioxide
- Water vapour
- Sulphur dioxide
- Tropospheric ozone

12. The ranking of countries as the highest per capita emitters of carbon dioxide in the world, takes into account their annual CO₂ emissions from

1. burning of fossil fuels.
2. cement manufacture.
3. land use such as deforestation.

Which of the statement(s) given above is/are correct?

- Only 1
- 1 and 2
- 1 and 3
- All of these

13. Below are the environmental benefits of organic farming

1. It aims to produce food while establishing an ecological balance to prevent soil fertility or pest problems.
2. It permits the use of GMOs.
3. It produces more biodiversity than other farming systems.
4. It improves the water retention capacity of the soil.

Which of the statements given above are correct?

- 1, 2 and 4
- 1 and 4
- 1, 3 and 4
- All of these